

Amogh Jayasimha

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Education

Northeastern University

Boston, MA

Master of Science in Information Systems, **GPA: 3.84**

Expected: Aug 2026

Relevant Courses: CSYE 6225 – Cloud Computing, TELE 6420 – Infrastructure Automation Design and Tools (Puppet, Ansible)

PES University

Bangalore, India

Bachelor of Technology in Electronics and Communication Engineering

Graduation Date: June 2022

Technical Skills

- **Languages and technologies:** Python (OOP, FastAPI, Flask, Boto3), Golang, Bash, PowerShell, JavaScript, Git, PostgreSQL, MySQL, Redis, Streamlit, Rust (learning); Linux/UNIX
- **Cloud & Platforms:** AWS (EC2, IAM, KMS, CloudWatch), GCP (GKE, Compute Engine, IAM, Ops Suite), OpenStack
- **Containers & Kubernetes:** Docker, Kubernetes, Helm, ArgoCD, Kargo, KEDA, Kubernetes NetworkPolicies, cert-manager, Chaos Mesh, Operators, Stateful services
- **CI/CD, GitOps & Security:** GitHub Actions, GitOps promotion pipelines, CI/CD optimization, Terraform, Cosign/Sigstore, Kyverno, External Secrets Operator
- **Observability & Reliability:** Prometheus, Grafana, Loki, CloudWatch, SLI/SLO monitoring, incident response and production debugging

Work Experience

Radius (EVM-based Blockchain Infrastructure)

Cambridge, MA

Platform / Infrastructure Engineering Co-op

Sep 2025 – Dec 2025

- Optimized PR deployment workflows by differentiating config-only vs. code changes using **registry-level image retagging**, cutting deployment time from **~17–27 min to ~2 min** while preserving isolation and correctness.
- Designed and implemented **default-deny Kubernetes NetworkPolicies** for EVM blockchain services, enforcing component-scoped ingress/egress rules, **DNS allowlists**, and explicit infrastructure access before production rollout.
- Led **on-call incident response** to chaos-testing failures by provisioning **persistent volumes** and **anti-affinity rules** for stateful services (storage layers, shard managers), eliminating data loss and restoring **service availability** during node rotation.
- Unified **Kargo** promotion pipeline (benchmarking, staging, testnet, mainnet) with **GitHub Deployments** and **ArgoCD** notifications for end-to-end traceability with Slack alerts and compare URLs.
- Improved **platform engineering** reliability by aligning CI and **Helm** workflows with operator defaults, reducing reliance on brittle **CRD**-required fields and eliminating a class of deployment failures in operator-managed workloads.

Societe Generale

Bangalore, India

Systems DevOps Engineer

July 2022 – July 2024

- Automated migration and decommissioning of **200+** legacy servers on the CTB IaaS transformation team (Retail, Corporate, Investment Banking) using Shell scripts, Python FastAPI, and **HP Server Automation (HPSA)**, reducing manual effort by **40%**.
- Built **GitOps pipelines** with **GitHub Actions** to auto-provision IaaS systems, cutting provisioning time from **2 hrs to 25 min**; authored **PowerShell** scripts for compliance checks and DNS remediation across **PCI-DSS** banking infrastructure.
- Developed **Python/Airflow DAG** pipelines to enforce **PCI-DSS** security compliance checks and eliminate **toil** in post-migration validation workflows across cloud perimeters.
- Built **capacity planning** tooling (Python, SQLite3, Plotly/Dash) to model server utilization; reduced cloud costs by **25%** via Boto3 right-sizing automation, earning **Spot Award** from Global Head.
- Created **Prometheus/Grafana** observability stack for **150+** cloud instances across AWS and GCP; supported datacenter migration planning and Windows Server troubleshooting during IaaS transformation.

Open Source Contributions

Kubernetes Supply Chain Security Platform

[GitHub](#)

Kyverno admission control enforcing cryptographic image signature verification (**Cosign/Sigstore**), **SBOM validation**, and **CVE-blocking** policies before containers reach production. Featured in the official Kyverno community showcase as a reference implementation of supply chain security.

Projects

k8s-chaos-promotion: Chaos-Gated GitOps Promotion Pipeline

[GitHub](#) — *In Development*

- GitOps pipeline on **GKE** where staging is hard-blocked from production until the app survives live **Chaos Mesh** fault injection (pod kills, network latency, RPC failures) under **k6** load via a **Kargo AnalysisTemplate** gate – **no prior public reference implementation**.
- Provisioned **GKE**, VPC-native networking, and GCS state via **Terraform** with fully keyless architecture (**Workload Identity Federation**, GitHub Actions OIDC); wired **ArgoCD** App-of-Apps, Kargo, ESO, and **Prometheus/Grafana/Loki** for full observability.
- Integrated **x402**-style HTTP micropayment flow with on-chain settlement via Radius EVM (web3.py, Foundry); validated with **~132 real on-chain transactions** across dev and staging.

Fully Automated Cloud-Native FastAPI Application on AWS & GCP

[GitHub](#) — Jan 2025 – Apr 2025

- Built shared **Packer** templates for AWS EC2/GCP; automated multi-cloud pipelines via **GitHub Actions** (OIDC) and **Terraform** with PostgreSQL, KMS, Secrets Manager, and Auto Scaling; configured **CloudWatch** & GCP Ops Suite monitoring with IAM least-privilege and **KMS encryption**.